ASSIGNMENT 1 – TASK 5: Apply Static Members and the Object Life Cycle

Bloom's Level: K4 – Analyse | Marks: 3

```
class TicketNumberGenerator {
    private static long next = 1000;

    public static synchronized long nextNumber() {
        return next++;
    }
}
```

The counter is static because ticket numbers must be shared across ticket objects. An instance counter would start separately for every generator/facility object and could produce duplicate ticket numbers. For two facilities in one Java process, a single static counter can be shared. For multiple servers or application restarts, use a database sequence or another durable central generator.

Static methods can be called through the class and cannot use instance fields directly without an object reference. A static block is suitable for one-time class initialisation, not for routine ticket creation. When a vehicle exits, close the ticket and remove active references from the facility. The ticket becomes eligible for garbage collection only when no live object can reach it; collection timing is not guaranteed.

For k facilities and n tickets, one shared static counter uses O(1) counter space; ticket objects and retained ticket history use O(n) space. Separate instance counters use O(k) counter storage and risk collisions unless ticket numbers include a facility identifier.